

DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I
WEEK 14: LECTURE NOTE

COMPLEX NUMBERS

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The Imaginary Unit

There are some equations that can not be solved using real numbers. For instance, what values of x satisfy the equation

$$x^2 + 1 = 0.$$

There is no real number that when multiplied by itself gives a negative number. Similarly, there is no real number x such that

$$x^2 = -9, \text{ or}$$

$$x^2 = -4, \text{ or}$$

$$x^2 = -8$$

To overcome this deficiency, mathematicians created an expanded system of numbers using the number called **the imaginary unit** i defined as

$$i = \sqrt{-1},$$

where

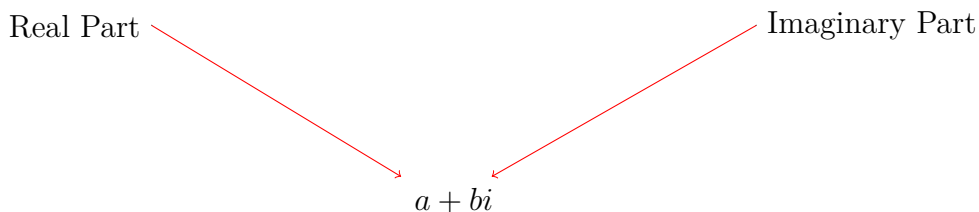
$$i^2 = -1.$$

Any equation like the ones above has at least one solution in the complex numbers. By adding real numbers to real multiples of this imaginary unit, we obtain the set of **complex numbers**.

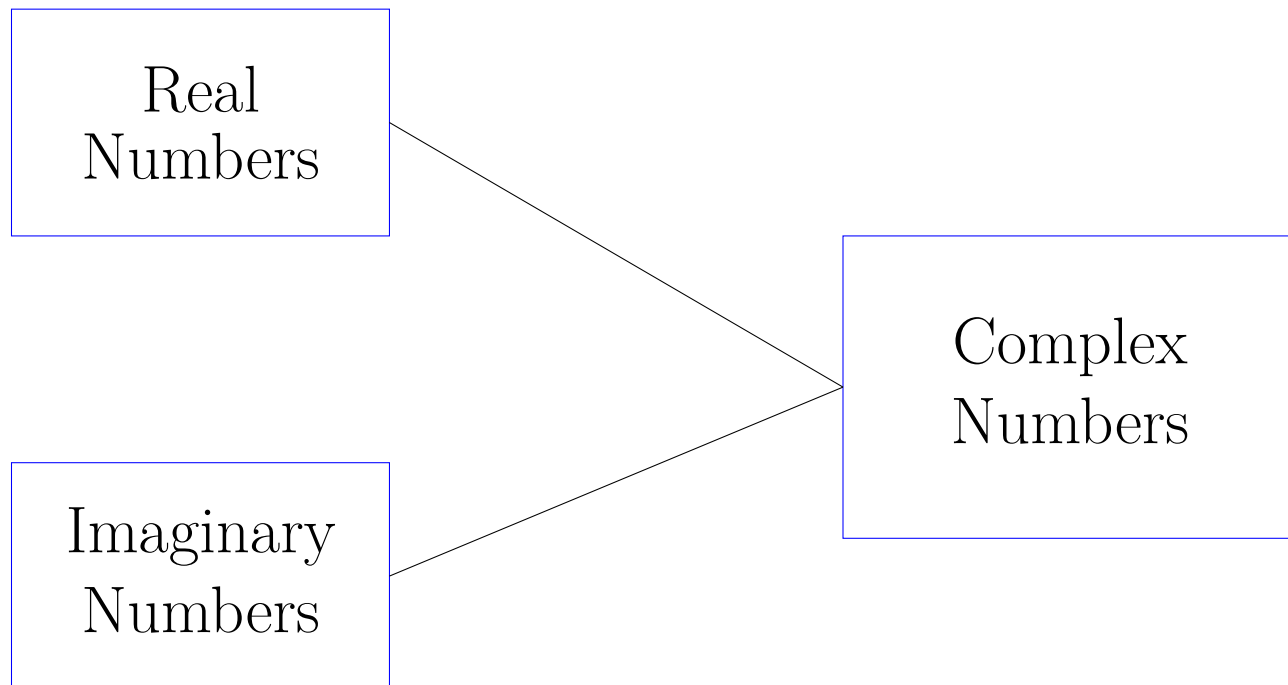
Definition 1. If a and b are real numbers, then the number $a+bi$ is a **complex number**, and it is said to be written in **standard form**. If $b = 0$, then the number $a + bi = a$ is a real number. If $b \neq 0$, then the number $a + bi$ is called an **imaginary number**. A number of the form bi , where $b \neq 0$, is called a **pure imaginary number**. The set of complex numbers, denoted by $\mathbb{C}$, is defined as

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}.$$

In the standard form $a+bi$, the real number a is called **the real part of the complex number** $a + bi$ and the number bi (where b is a real number) is called **the imaginary part of the complex number**.



The set of real numbers is a subset of the set of complex numbers because for every real number a we can write $a = a + 0i$.



Definition 2 (Equality of Complex Numbers). *Two complex numbers $a + bi$ and $c + di$, written in standard form, are equal to each other*

$$a + bi = c + di$$

if and only if $a = c$ and $b = d$.

OPERATIONS WITH COMPLEX NUMBERS

Definition 3. *Let $z = a + bi$ and $w = c + di$ be two complex numbers written in standard form.*

1. Addition

$$z + w = (a + c) + (b + d)i$$

2. Subtraction

$$z - w = (a - c) + (b - d)i$$

3. Multiplication

$$zw = (a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

4. **Division.** For $c \neq 0$ or $d \neq 0$, that is, $w \neq 0$, we have

$$\begin{aligned} z \div w &= \frac{a + bi}{c + di} \\ &= \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i. \end{aligned}$$

The usual rules of arithmetic apply to complex numbers. For instance, if z , w and u are complex numbers, then

- Associativity

$$(z + w) + u = z + (w + u)$$

Addition is associative

$$(zw)u = z(wu)$$

Multiplication is associative

- Commutativity

$$z + w = w + z$$

Addition is commutative

$$zw = wz$$

Multiplication is commutative

- Distributivity

$$z(w + u) + u = zw + zu$$

Multiplication is distributive over Addition

Definition 4 (Complex Conjugates). Let $z = a + bi$ be the standard form of the complex number z . Then the complex conjugate of z , denoted by $\bar{z}$, is defined as

$$\bar{z} = a - bi.$$

Proposition 5 (PROPERTIES OF THE COMPLEX CONJUGATE). For a complex number $z = a + bi$ and its conjugate $\bar{z} = a - bi$, the following properties hold: I

1. $Re(\bar{z}) = Re(z)$
2. $Im(\bar{z}) = -Im(z)$.
3. If z is real ($z = a$), then $\bar{z} = z$.
4. If z is purely imaginary ($z = bi$), then $\bar{z} = -z$.
5. $z + \bar{z} = 2Re(z) = 2a$
6. $z - \bar{z} = 2Im(z)i = 2bi$.
7. $z\bar{z} = (a + bi)(a - bi) = a^2 - (bi)^2 = a^2 - (-b^2) = a^2 + b^2$.
8. $\bar{\bar{z}} = z$.

Example 6 (Addition and subtraction of complex numbers). Perform the operation(s) and write the result in standard form.

1. $(3 - i) + (2 + 3i) = (3 + 2) + (-1 + 3)i = \mathbf{5 + 2i}$

$$2. \sqrt{-4} + (-4 - \sqrt{-4}) = 2i - 4 - 2i = (0 - 4) + (2 - 2)i = -4 + 0i = -4$$

$$3. 4 + (-2 + 2i) - (3 - 5i) = (4 - 2 - 3) + (0 + 2 + 5)i = -1 + 7i.$$

$$4. \sqrt{-4} + (-3 - \sqrt{-9}) = (0 + 2i) + (-3 - 3i) = (0 - 3) + (2 - 3)i = -3 - i.$$

Example 7. Perform the operation(s) and write the result in standard form.

$$1. \sqrt{-4}\sqrt{-16} = (2i)(4i) = 8i^2 = -8.$$

$$2. (2 - 3i)(4 + 3i) = [2(4) - (-3)(3)] + [2(3) + (-3)(4)]i = 17 - 6i.$$

$$3. (4 + 3i)^2 = 4^2 + 2(4)(3i) + (3i)^2 = 16 + 24i - 9 = 7 + 24i.$$

$$4. ((1 + i)(1 - i)) = 1^2 + 1^2 = 1 + 1 = 2$$

$$5. 3i(-1 + 5i) = -3i + 15i^2 = -3i - 15 = -15 - 3i.$$

Example 8. Write the quotient $\frac{2 + 3i}{4 - 2i}$ in standard form.

Solution.

$$\begin{aligned} \frac{2 + 3i}{4 - 2i} &= \frac{2 + 3i}{4 - 2i} \cdot \frac{4 + 2i}{4 + 2i} \\ &= \frac{(8 + 6) + (12 - 4)i}{4^2 + 2^2} \\ &= \frac{14}{20} + \frac{6}{20}i \\ &= \frac{7}{10} + \frac{3}{10}i \end{aligned}$$

Example 9. Write the quotient $\frac{8 - 7i}{1 + 2i}$ in standard form.

Solution.

$$\begin{aligned} \frac{8 - 7i}{1 + 2i} &= \frac{8 - 7i}{1 + 2i} \cdot \frac{1 - 2i}{1 - 2i} \\ &= \frac{(8 - 14) + (-7 - 16)i}{1^2 + 2^2} \\ &= \frac{-6}{5} + \frac{-23}{5}i \\ &= -\frac{6}{5} - \frac{23}{5}i \end{aligned}$$

Example 10. Perform the operations for

$$\frac{2}{1 + i} - \frac{3}{1 - i}$$

and write in standard form.

$$\begin{aligned} \frac{2}{1 + i} - \frac{3}{1 - i} &= \frac{2(1 - i) - 3(1 + i)}{(1 + i)(1 - i)} \\ &= \frac{2 - 2i - 3 - 3i}{1^2 + 1^2} \\ &= \frac{-1 - 5i}{2} \\ &= -\frac{1}{2} - \frac{5}{2}i \end{aligned}$$